BIRZEIT UNIVERSITY

COMP2310 – Advanced Programming

Assignment# 2

Objectives:

1. Create Classes and Objects.
2. Process Array of objects.
3. To use the **String** class to process immutable strings.

Specification

Submission: **Online through ITC**.

What to submit: Your **own** **well-structured** and **well-commented** JAVAfiles (**.java**) and PDF file

Deadline: **8/5/2025** by 8pm. (The online submission will be disabled after this time).

Important

- Academic **honesty**:
  - This is an individual assignment. Individual assignments must be each student's own work.
  - Copying 1 line from a friend or the internet will be considered cheating.
  - Cheating will result in an official university disciplinary review and the University regulations will be strictly enforced.

Task1: Passenger Class

A **Passenger** class has the following attributes:

- First name (Letters only and at least 3 letters)
- Last name(Letters only and at least 3 letters)
- Gender (M/F)
- Passport number (numbers only and 7 digits)
- Nationality
- Has an entry VISA
- A static parameter for the number of created passenger objects.

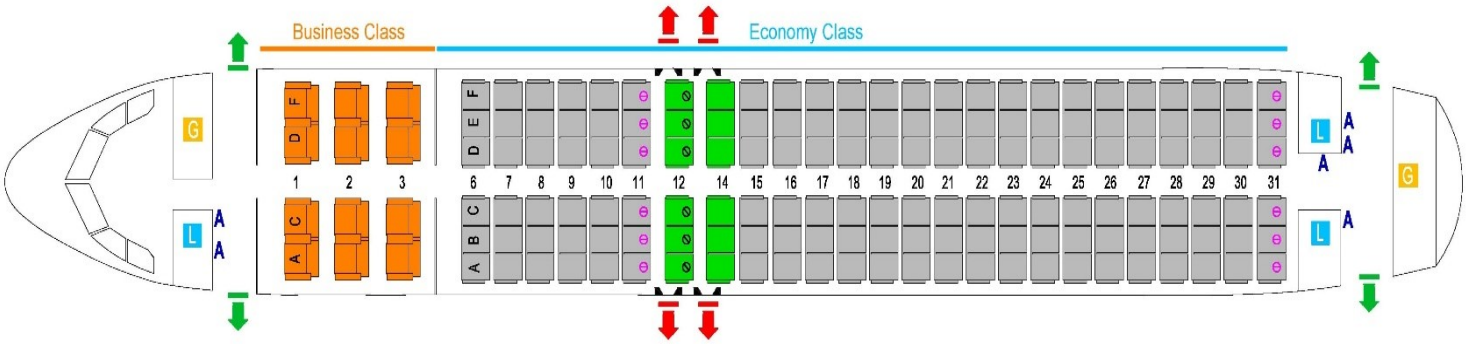
And the following behaviours:

- Setters and getters to all the above class attributes
- One constructor using all the instance attributes.
- One method call printPassengerInfo to print the passenger information.

1. Draw the UML diagram for the Passenger class. (use any free online tool to draw UML and save it as a PDF file. e.g. https://www.draw.io/)
2. Implement the class.

Task2: A320 Flight Seat Reservation System

The flowing figure shows the A320 flight seat Configuration:



- In this task, you need to create a 2D **ragged array** of Passenger seats with a length of 31 rows. The first 3 rows are reserved for the Business class seats as shown in the above figure. The rows from 6 to 31 are reserved for the Economy class seats. Rows 4, 5, and 13 has no seats. The seat number is consist of one character (A to F) and a number (1 - 31) (e.g. F17).

- Create a driver class that has the following:
  - At the beginning of execution, create and initialize the above mentionedarray (i.e. flight is empty of passengers, all seats are available).
  - In an infinite loopshow the following menu options:
    0) to reserve a random empty seat (business or economy) for a new passenger.
    1) to reserve a new seat (business or economy) for a new passenger (if the provided seat number is empty)
    2) to display the passenger information for a specific seat (if the seat is not empty)
    3) to search for a passenger by first or last name, if found return the seat number.
    4) to search for passenger by passport number, if found return the seat number.
    5) to print a list of the passengers and there seat numbers.
    6) to exit execution
  - Adequately implement the previous options (from 0 to 6)

Good Luck!